

Beechui (Katy) Koo, MS

koobcbc@gmail.com | (617) 888-0042 | Chicago, IL
koobcbc.github.io | linkedin.com/in/katybkoo | [google scholar](https://scholar.google.com/citations?user=koobcbc)

EDUCATION

University of Chicago, Chicago, IL 09/2024 – 12/2025
 M.S. in Applied Data Science (GPA: 3.98/4.00)

- Relevant Courses: Data Engineering, Statistical Models for Data Science, Machine Learning I/II, Bayesian Machine Learning with GenAI Applications, Time Series Analysis and Forecasting, Leadership and Consulting, Computer Vision, GenAI Principles & Applications, Capstone I/II

Boston College, Brighton, MA 08/2017 – 05/2021
 B.S. in Computer Science, Minor in Biology

- Relevant Courses: General Chemistry I/II, Organic Chemistry I/II, Cell Biology, Genomics, Computer Science I/II, Algorithms, Object-Oriented Design, Computer Systems, Intro to Physics I (Calc) w/ Lab, Multivariable Calculus, AP Physics B1/B2/C Equivalent

University of California San Diego, Online 06/2025 – 08/2025
 Accredited Physics Extended Courses

- Relevant Courses: Mechanics and Thermodynamics, Electricity and Magnetism, Modern Physics

RESEARCH EXPERIENCE / PROJECTS

University of Chicago Medicine (Advisor: Dr. James J. Sohn) 02/2025 – Current
Graduate Research Associate

- Performed quantitative evaluation of commercial CT metal artifact reduction (MAR) algorithms, benchmarking CT number accuracy, streak suppression, and residual artifact profiles relevant to radiation treatment planning.
- Developed TOPAS Monte Carlo simulations incorporating high-Z nanomaterials to model primary and scattered photon attenuation, optimizing vault shielding thickness and material composition for next-generation radiation therapy facilities.
- Simulated the dosimetric impact of tattoo pigments on patient skin by modeling spectral-dependent photon absorption and scatter using TOPAS-based Monte Carlo methods, quantifying surface dose enhancement across beam energies and pigment compositions.
- Designed and implemented an automated pipeline in Python/Rhino for generating variable-thickness 3D printed molds tailored for conformal surface brachytherapy, improving patient-specific fit and dose conformity.

University of Chicago, Data Science Institute (Advisor: Prof. Nick Kadochnikov) 02/2025 – 12/2025
Capstone Partner AI Engineer

- Designed and implemented an AI-driven diagnostic assistant that analyzes user-uploaded skin or dental photos to identify potential diseases.
- Integrated CNN-based computer vision models with Gemini 2.5 LLM to provide explainable condition summaries and recommend next steps using verified medical sources.
- Developed a cross-platform mobile application using React Native (Expo Go), with backend integration into Firestore database, Google BigQuery, and Vertex AI for scalable data storage and inference.

United Airlines GenAI Hackathon 06/2025

- Led the development of an agent-to-agent GenAI solution for real-time airline fault reporting, winning 2nd place among presenting groups, seamlessly operable across iOS, Android, and web applications.
- Architected a modular multi-agent system using LangChain and Gemini to handle multimodal inputs and automate issue triage, significantly improving response time and accuracy.
- Trained and integrated specialized AI agents (Image, Form, Submission, Supervisor) to classify images, auto-fill structured forms, and execute Retrieval-Augmented Generation (RAG)-driven fault resolution pipelines.

University of Chicago, Data Science Institute (Advisor: Dr. Utku Pamuksuz) 11/2024 – 05/2025
Machine Learning Project

- Built a machine-learning pipeline to predict BMI from images by benchmarking CNN-based models, and deploy the best model in a live smartphone app. (04/2025 – 05/2025)
- Developed a machine learning framework integrating structured and unstructured movie data (metadata, reviews, and plots) to predict box office revenue using XGBoost, NLP-based sentiment analysis (VADER, XLM-RoBERTa), and advanced feature engineering for cast and crew metrics. (02/2025 – 04/2025)
- Constructed an end-to-end deep-learning pipeline that estimates left-ventricular ejection fraction from echocardiogram videos

on public echo datasets. (02/2025 – 03/2025)

- Implemented a stock price prediction model by performing sentiment classification on financial news headlines using a BERT-based NLP pipeline, and integrated results into an LSTM model to forecast stock price movements. (11/2024 – 01/2025)

Harvard Medical School, Boston Children's Hospital (Advisor: Dr. Sun Hur)

05/2019 – 05/2020

Undergraduate Research Assistant

- Studied the molecular mechanisms that drive stress granule assembly in response to double-stranded RNA and how these RNA–protein condensates influence post-transcriptional gene regulation during antiviral stress.
- Performed dsRNA transfection assays (e.g. 162 bp 5'-ppp dsRNA) in mammalian U2OS cells to induce stress granule formation and triggered innate immune response, in order to study the upstream triggers of RNA–protein condensate assembly.
- Used confocal fluorescence microscopy to visualize co-localization of canonical stress granule marker G3BP1 with RIG-I, MAVS, and dsRNA, and quantified granule frequency/size via image analysis across >200 granules per condition.
- Conducted Western blot analyses to assess activation of innate immune signaling, including IRF3 phosphorylation, TBK1 activation, and MAVS-mediated signaling downstream of stress granule formation.

PROFESSIONAL EXPERIENCE

Sohn Lab, Dept. of Radiation and Cellular Oncology, University of Chicago Medicine, Chicago, IL

02/2025 – Current

Research Associate

- Led independent research projects on “Quantitative Comparison of CT Metal Artifact Reduction (MAR) Algorithms in Radiation Oncology” and “TOPAS Monte Carlo Simulations with Cutting-Edge Nanomaterial Applications to Optimize Shielding Designs for Modern Radiation Therapy Facilities,” etc.
- Directly managed a team of interns and junior researchers through weekly meetings, milestone planning, and technical lectures on applied medical physics and computational modeling.
- Established project roadmaps and analysis pipelines, including custom evaluation metrics (e.g., HU error maps, M-Error Index) and artifact quantification methods.
- Served as the primary liaison between the research team and faculty advisor, ensuring scientific alignment, resource coordination, and progress reporting.

Oncosoft Inc., Chicago, IL

06/2025 – 09/2025

Data Scientist / AI Engineer

- Designed and integrated an AI-powered clinical chatbot leveraging Retrieval-Augmented Generation (RAG) and specialized agents to provide real-time, context-aware responses by querying graph databases, LLMs (GPT-4), and external medical knowledge sources, contributing to a 200% increase in annual revenue by streamlining patient information access.
- Developed a full-stack clinical software platform using React.js and Django, enabling clinicians to view dashboards, analyze patient charts, create custom forms, and manage records efficiently for radiation therapy.
- Represented the platform at professional conferences, demonstrating features and engaging with potential users and stakeholders.

Xspha Biosciences Inc., Boston, MA

09/2021 – 07/2024

Software Engineer

- Developed an end-to-end multi-omics data pipeline for immuno-oncology analysis, integrating Python ML frameworks with MySQL and AWS S3 to automate ingestion, preprocessing, and quality-controlled analysis.
- Led the development of a cloud-based single-cell flow cytometry analysis platform built with Django and React.js, integrating ML classifiers for immune cell population profiling and deploying scalable data processing pipelines using AWS EC2.
- Replaced manual Excel-based plotting by building interactive dashboards with Plotly Dash, custom JavaScript callbacks, and Django back-end, reducing result-check steps and centralizing insights, directly reducing validation time by 10x.

Hur Lab, Harvard Medical School, Boston Children's Hospital, Boston, MA

05/2019 – 05/2020

Research Assistant

- Contributed to research investigating innate immune responses to viral infection, with a focus on the RIG-I/MAVS signaling pathway and downstream cellular mechanisms.
- Assisted in experimental design, execution, and data interpretation for molecular and cell biology studies aimed at elucidating antiviral defense mechanisms.
- Performed core laboratory techniques including molecular cloning, mammalian cell culture and transfection, SDS-PAGE, western blotting, cDNA synthesis, and fluorescence microscopy.

PUBLICATIONS

1. **Beechui Koo**, Hyunuk Jung, Mitchell Polizzi, Indra J. Das, Siyong Kim, James J. Sohn. “*Quantitative Comparison of CT Metal Artifact Reduction (MAR) Algorithms in Radiation Oncology*” Radiation Physics and Chemistry. (Minor Revision Submitted)

2. Max Paget, Cristhian Cadena, Sadeem Ahmad, Hai-Tao Wang, Tristan X Jordan, Ehyun Kim, **Beechui Koo**, Shawn M Lyons, Pavel Ivanov, Xin Mu, Sun Hur. “*Stress granules are shock absorbers that prevent excessive innate immune responses to dsRNA*” Mol Cell. 2023;83(7):1180-1196.e8.

3. **Beechui Koo**, Richard Xu, Morgan Glennie, Jungwook Park, James J. Sohn. “*TOPAS Monte Carlo Simulations with Cutting-Edge Nanomaterial Applications to Optimize Shielding Designs for Modern Radiation Therapy Facilities*” Med. Phys. (In Preparation; Manuscript in Revision)

4. Hongjun Park, **Beechui Koo**, Jungwook Park, Siyong Kim, James J. Sohn. “*Does Tattoo Ink Alter Radiation Dose in Human Skin?*” Med. Phys. (In Preparation; Manuscript in Revision)

5. **Beechui Koo**, James J. Sohn, Jason W. Sohn. “*Automated Design of Variable Thickness 3D Printed Molds for Conformal Surface Brachytherapy*” Med. Phys. (In Preparation)

CONFERENCE ABSTRACTS

UChicagoGRAD 7th Annual Transcending Boundaries Research Symposium, Chicago, IL, USA (2025). (Oral)

US-Korea Conference, Atlanta, GA, USA (2025). (Poster)

RSNA Radiological Society of North America (RSNA), Chicago, IL, USA (2025). (Poster)

CERTIFICATION

MIT Data Science & Machine Learning 12-week Certificate (MIT-Schwarzman College of Computing)	05/2025
General Assembly Software Engineering 14-week Immersive Course Certificate	08/2020

HONORS AND SCHOLARSHIPS

Korean Honor Scholarship 2025 (\$1500), Embassy of Republic of Korea in the USA	08/2025
United Airlines GenAI Hackathon 2 nd Place	06/2025
The University of Chicago Data Science Institute Merit Scholarship (\$10000)	08/2024

AFFILIATIONS

Member, American Association of Physicists in Medicine (AAPM), 2025–Present

Member, Radiological Society of North America (RSNA), 2025–Present

Member, Korean-American Scientists and Engineers Association (KSEA), 2025–Present

Member, University of Chicago Korean Graduate Student Association (KGSa), 2024–Present

Co-President, Quiet Waters Boston College, 2018–2021

SKILLS

Programming: Python, R, Django, React.js, Ruby on Rails, SQL (MySQL, PostgreSQL), NoSQL (MongoDB/Mongoose)

AI & Machine Learning: Generative AI (LLMs, RAG, LangChain), PyTorch, TensorFlow, Keras, Scikit-learn, Computer Vision, QLoRA

Cloud & DevOps: AWS, GCP

Data Science & Analytics: Data Pipelines (cleaning, wrangling, visualization, modeling, interpretation), Statistical Analysis, Hypothesis Testing, Tableau, Pandas, OpenCV

Monte-Carlo simulation: TOPAS

CAD/Modeling Software/3D Printing: AutoCAD, Fusion 360, Rhino, Bambu Studio, Makerspace, 3D printing

REFERENCES

James J. Sohn, PhD., Assistant Professor Dept. of Radiation and Cellular Oncology The University of Chicago Chicago, IL 60637 Email: Jooyoung.sohn@bsd.uchicago.edu	Sun Hur, PhD., Oscar M. Schloss, MD Professor of Pediatrics Dept. of Biological Chemistry and Molecular Pharmacology Harvard Medical School Boston, MA 02115 Email: sun.hur@crystal.harvard.edu
---	---

Jason W. Sohn, PhD., Director of Medical Physics
Dept. of Radiation Oncology
Beth Israel Deaconess Medical Center
Boston, MA 02215
Email: jsohn3@bidmc.harvard.edu

Nick Kadochnikov, Assistant Clinical Professor
Data Science Institute
The University of Chicago
Chicago, IL 60637
Email: kadochnikov@uchicago.edu

Utku Pamuksuz, PhD., Associate Clinical Professor
Data Science Institute
The University of Chicago
Chicago, IL 60637
Email: pamuksuz@uchicago.edu